

Flanker Task EEG functional data analysis

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Introduction

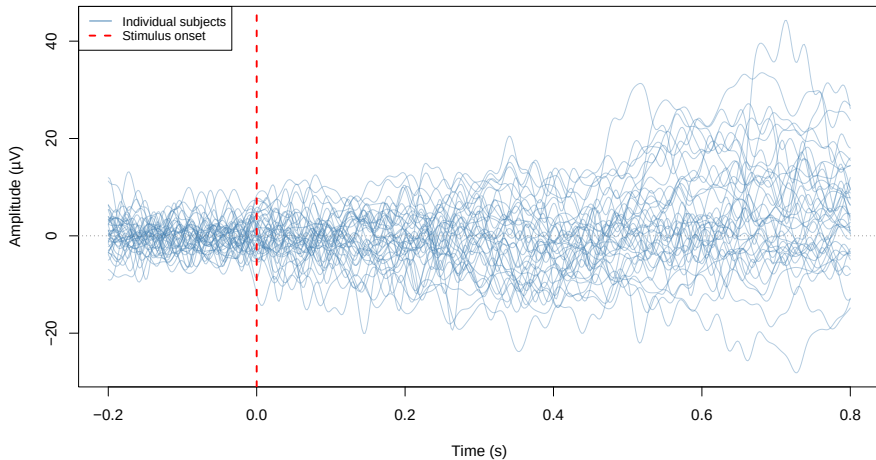
Project

- This dataset contains:
 - EEG brain recordings from 127 young adults.
 - Retrospective objective and subjective reports of childhood family socioeconomic status (SES).
 - SES indicators in adulthood.
 - Self-reports of ADHD symptoms.
- This project applies Functional Data Analysis methods to EEG data from 39 adults recorded during a Flanker task at electrode channel FC1.
- Data source - Cognitive Electrophysiology in Socioeconomic Context dataset (doi:10.18112/openneuro.ds006018.v1.2.2).

Data

Data

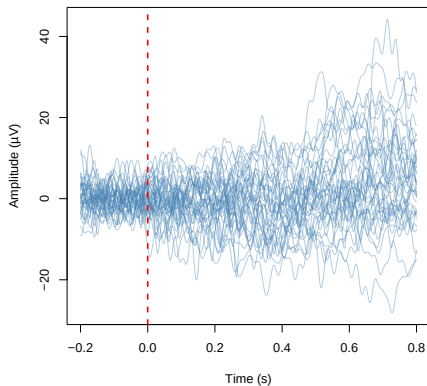
Flanker Task — Channel FC1 — All 39 Subjects



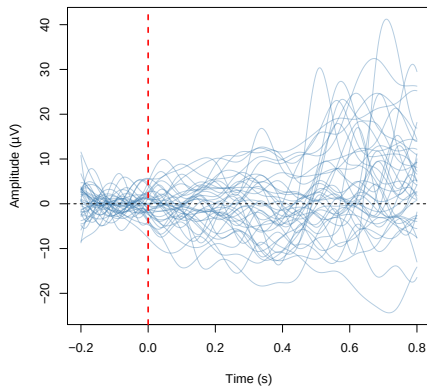
Smoothing

Smoothed data

Before Smoothing (raw)

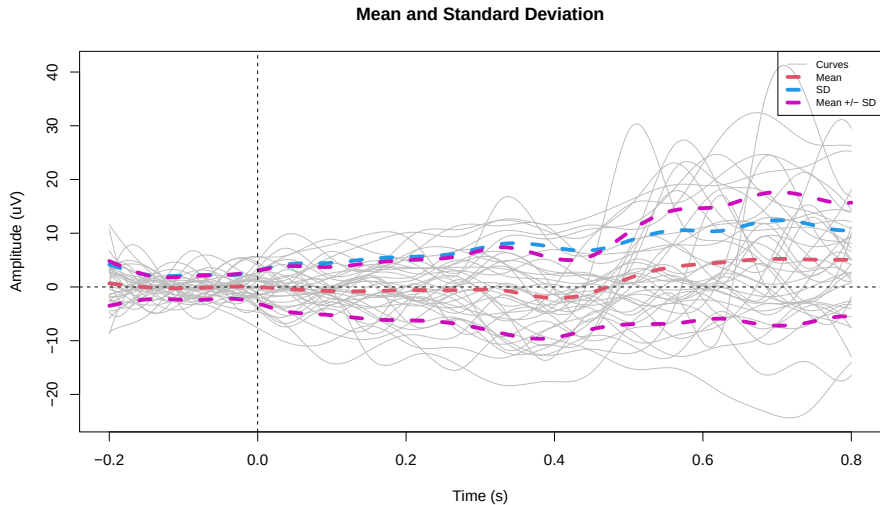


After Smoothing (lambda = 1e-05)

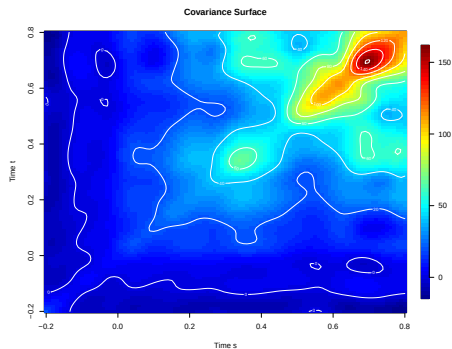
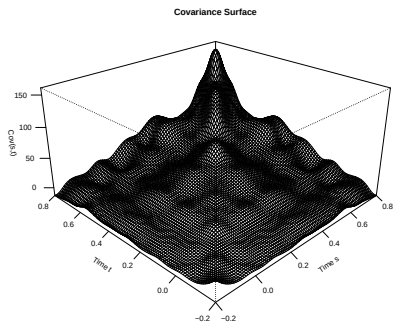


Exploratory Data Analysis

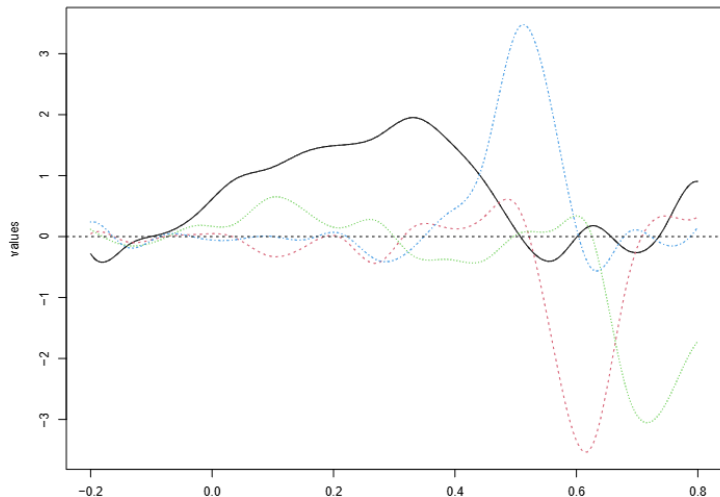
Mean and standard deviation



Covariance

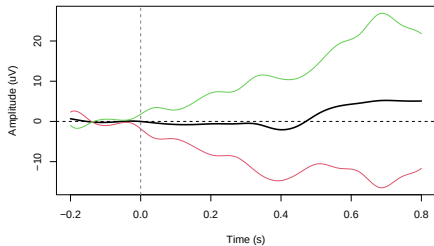


Principal Component Analysis I

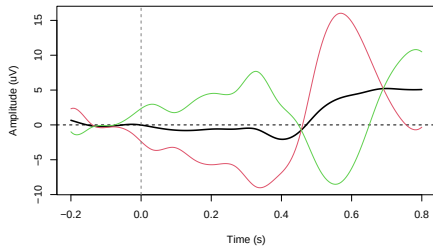


Principal Component Analysis II

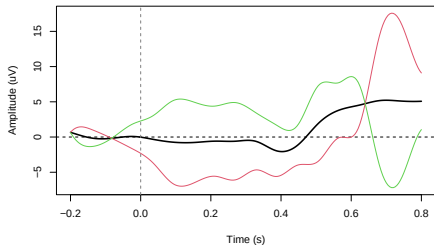
PC1 (61.1%)



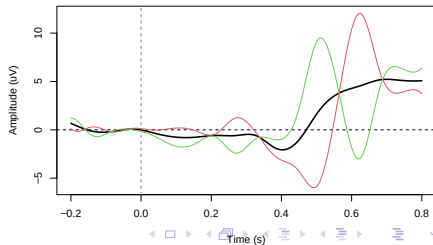
PC2 (13.6%)



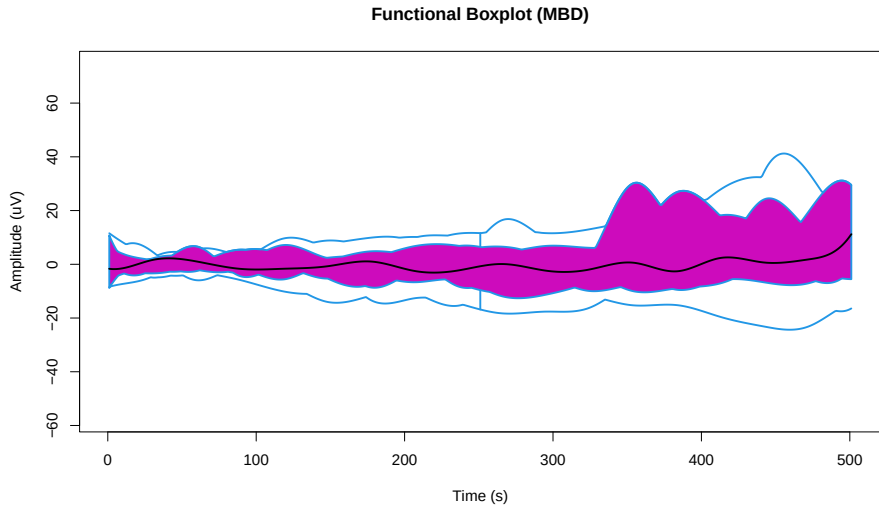
PC3 (11.9%)



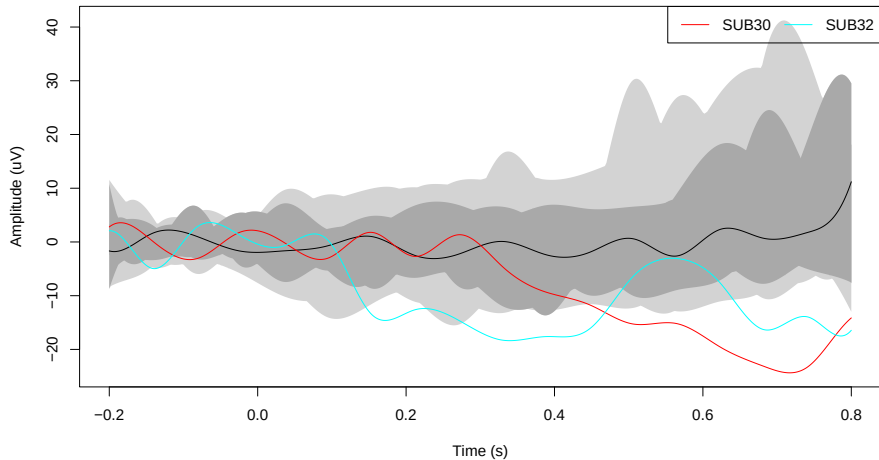
PC4 (3.3%)



Functional Box Plot



Outliers



Thank you for your time!